B.TECH SEMESTER EXAMINATION - 2022
Department of Computer Science & Engineering
Subject: Operating Systems (CS-401)
Time: 3 Hours Max. Marks: 100

Note: Answer any five questions.

1. What is an Operating System? Explain its main functions and types. [10 Marks]

2. Explain process state transitions with a neat diagram. What information is stored in a PCB? [10 Marks]

3. Define CPU scheduling. Compare FCFS, Round Robin, and SJF scheduling algorithms with an example. [15 Marks]

4. What is a Deadlock? Explain the four necessary conditions for deadlock occurrence. [10 Marks]

5. Explain the concept of Virtual Memory. What is demand paging? [10 Marks]

6. Differentiate between paging and segmentation with suitable diagrams. [15 Marks]

7. Write short notes on: (a) Thrashing (b) Semaphore (c) Critical Section Problem. [15 Marks]